

Joshua Williams

Alternative Energy and Power Generation Engineering | Penn State | Class of 2027
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EDUCATION

B.S. in Engineering, Alternative Energy and Power Generation

Aug 2023 - May 2027

Interdisciplinary coursework spanning Mechanical engineering and Electrical engineering.

Minor: Engineering Leadership

Penn State University: Hazleton: GPA: 3.50 - Honors Program, Tri-Alpha Honor Society

Coursework: (Mechanical & Electrical Engineering): Power Electronics, Solar Energy System Design, Fuel Cells and Electrochemistry, Strength of Materials, Thermodynamics II, Renewable Energy Systems, Modeling (Ansys/MATLAB)

TECHNICAL SKILLS

Software and Tools: AutoCAD | Fusion 360 | SolidWorks | MATLAB (Simulation) | Ansys (FEA/Simulation) | Arduino | Python | Microsoft Excel | 3D Printing / Rapid Prototyping

Engineering: CAD Modeling, RCA, BOM, Systems Integration, Mechanical Design, Power Electronics, Energy Conversion, Process Improvement, Quality Control, SOP Writing, LOTO Compliance

Programming: C/C++ (Arduino, ESP32), Python (Raspberry Pi)

PROFESSIONAL EXPERIENCE

Engineering Intern, Insteel Wire Products: Hazleton, PA

May 2025 - Present

- Executed a complete engineering cycle on production components: failure analysis, CAD redesign in Fusion 360, physical testing, reducing costs by up to 97% and generating over \$10,000 in annual savings.
- Produced 100+ 2D technical drawings in AutoCAD and developed 75+ 3D CAD components.
- Conducted FMEA on manufacturing processes, identifying failure modes and implementing corrective actions that reduced part defects and failures by over 50%.
- Authored a Quality Control Training Guide and revised SOPs and LOTO procedures to standardize testing protocols, reduce operator errors, and maintain safety compliance.
- Proposed and built out the plant's additive manufacturing program, growing it from zero to three 3D printers in daily production use, and completed a plant-wide drainage review to support future infrastructure planning.

Manager, Broyan's Farm Market: Nescopeck, PA

December 2020 - August 2025

- Supervised and trained a team of 10+ employees over four years, managing daily operations and inventory control.

ENGINEERING PROJECTS

Autonomous Robotic Rover: Hardware Lead

May 2026 - Present

- Designing a **large-scale autonomous 4WD robotic rover** in Fusion 360, including the chassis, suspension, steering, and custom 3D-printed components.
- Integrating a Raspberry Pi 4, ESP32, encoder-equipped motors, BTS7960 motor drivers, and a custom 4S4P battery system while leading hardware assembly and testing.
- Collaborating on autonomous navigation and developed project documentation, CAD, BOM, and hardware validation.

Single-Axis Solar Tracking System

Aug 2025 - May 2026

- Designed, built, and tested a single-axis solar tracker that delivered a validated 17.5% improvement in energy yield over a fixed-angle baseline, confirmed through controlled electrical output measurements at multiple angles.
- Awarded 1st Place, Spring 2026 Penn State Honors Research Symposium.

Far Field Sound Direction-of-Arrival Detector, Mechanical & Electrical Lead

Aug 2026 - Present

- Built a three-microphone array on a Teensy 4.0 that detects a sound and calculates its direction of arrival via cross-correlation, achieving accuracy within about a degree at a 10-foot test distance.

HONORS, AWARDS & PUBLICATIONS

- 1st Place - Spring 2026 Penn State Honors Research Symposium | Single-Axis Solar Tracking System
- Co-author, **"WIP: Workshop Activities Focused on Hands-On Abilities for Engineering Students."** Presented at the ASEE Annual Conference

LEADERSHIP

President, Science and Engineering Club, Penn State Hazleton:

Aug 2025 - Present

- Directed a series of technical projects including a PCB design workshop, Arduino motion sensor builds, and a drone build integrating custom PCBs and 3D-printed components; organized the annual Cardboard Boat Race engineering challenge. Currently leading the design and construction of a campus microgrid integrating 200–300W of solar, a 400W wind turbine, battery storage, and pumped hydro energy storage.